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NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Seventh Semester B.E. (E&C) (Credit System) Degree Examinations
 Supplementary Examinations – December 2025

21ECE214/20ECE44 – EMBEDDED SYSTEMS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	What is an embedded system? List any five applications of Embedded Systems and explain Digital camera with functional block diagram.	10	L*2	1	1
	b)	i) Compare the RISC and CISC Architecture ii) Briefly explain software development process	10	L2	1	1
2.	a)	What is a design metric? Explain the various commonly used design metrics.	10	L2	1	1
	b)	Design a Juice vending machine, cost of Juice Rs 75/- glass and the machine accepts quarters only. Draw a black-box view, come up with a state diagram and state table, minimize the logic, and then draw the final circuit.	10	L4	1	1
3.	a)	Explain the role of sensors and actuators in embedded system design. Illustrate each with one example.	10	L2	2	1
	b)	List the different external communication interfaces and explain any three interfaces in brief.	10	L2	2	1
		Unit – II				
4.	a)	Explain the various electronic control units used in the automotive embedded industry. Also explain the different communication buses used in automotive application.	10	L2	3	1
	b)	Briefly explain the fundamental issues in hardware software co-design.	10	L2	3	1
5.	a)	Explain the following; DFG model, DCFG model	10	L2	3	1
	b)	Design and explain an automatic tea/coffee vending machine based on FSM model for the following requirement: Tea/Coffee vending is initiated by user inserting a 5 rupee coin. After inserting the coin, the user can either select or 'Tea' or press 'Cancel' to cancel the order and take back the coin.	10	L4	3	1
6.	a)	Explain the super loop based embedded firmware design approach, with the help of execution flow and C program code.	10	L3	4	1
	b)	Explain the role of programming languages in system design; and also explain the process of conversion from assembly language to machine language.	10	L2	4	1
		Unit – III				
7.	a)	Explain the role of Integrated Development Environment (IDE) for embedded software development.	10	L2	5	1
	b)	What is cross compilation? Explain in detail the need for cross compilation.	10	L2	5	1
8.	a)	Explain the advantages and limitations of simulator based debugging.	10	L2	5	1
	b)	List and explain the various hardware debugging tools used in embedded product development.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
